

The background is a vibrant, stylized space scene. It features large, flowing, organic shapes in shades of red, purple, and blue. Scattered throughout are numerous small white stars and several planets. In the top right, there's a large red planet with orange and yellow patterns, and a smaller yellow and orange striped planet below it. In the bottom left, a yellow and orange striped planet is visible. Other planets in shades of blue and purple are scattered across the scene.

AI30

Simulation de conquête spatiale

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The background is a vibrant, abstract space-themed illustration. It features a gradient of colors from deep red on the left to light blue on the right. Scattered throughout are various celestial bodies: a planet with blue and white stripes in the top left, a planet with orange and white stripes in the top right, a planet with yellow and red stripes and a red ring in the bottom right, and several smaller planets with blue and white patterns. Numerous small white stars and sparkles are also visible.

01

Présentation du projet



Motivations

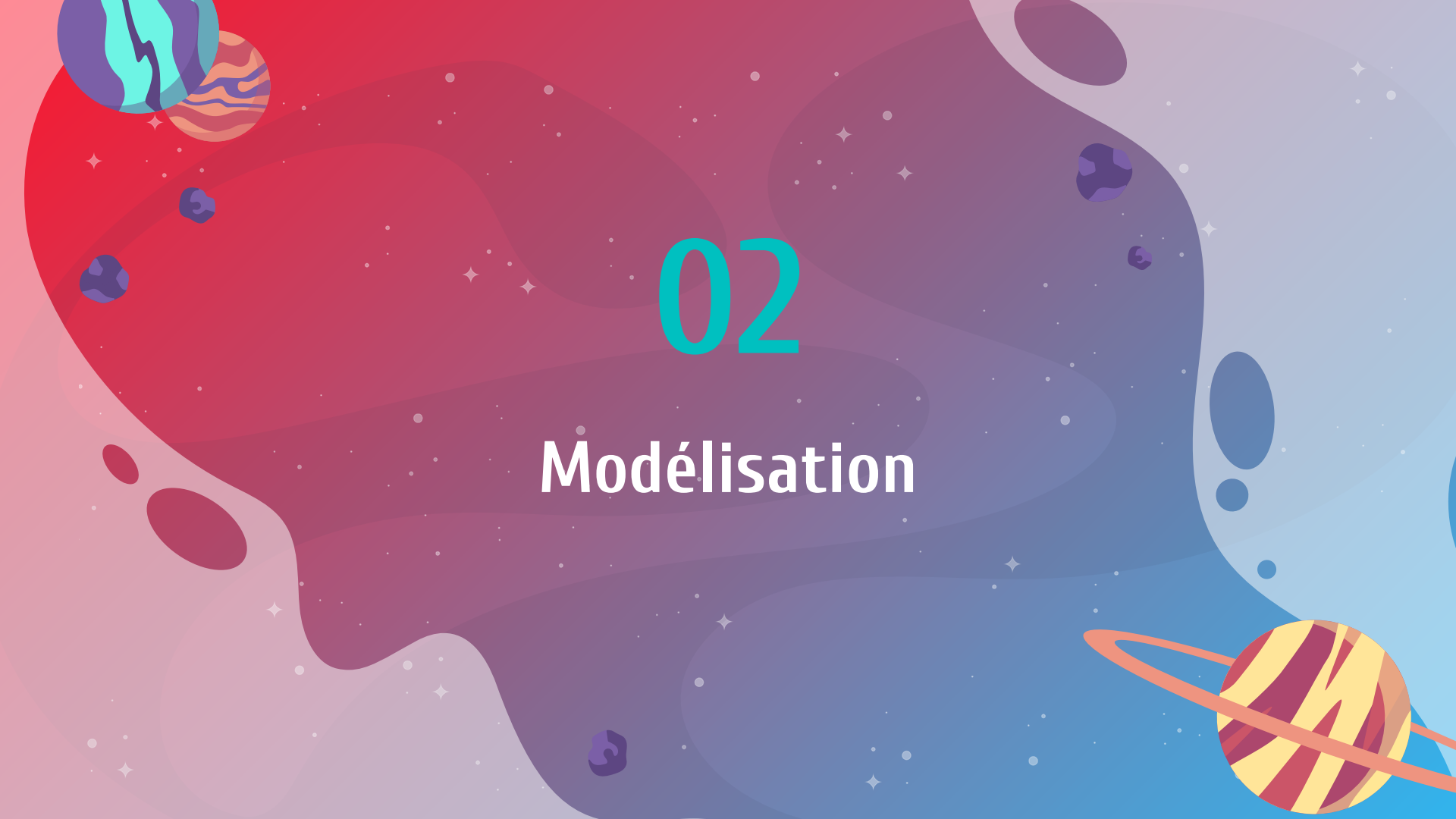




Problématique

Quelle est la meilleure stratégie qu'un pays puisse adopter pour conquérir le système solaire ?



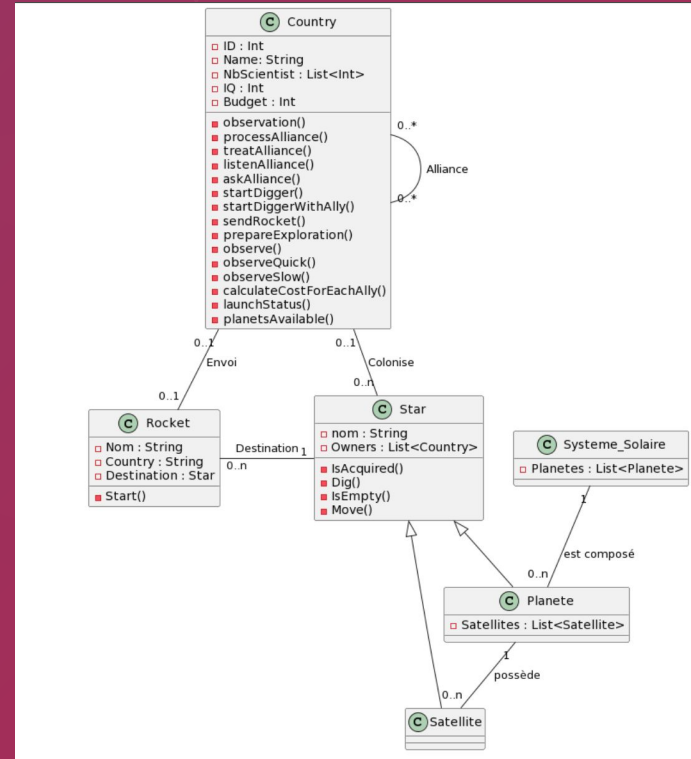
The background is a vibrant, abstract representation of space. It features a gradient from deep red on the left to light blue on the right. Scattered throughout are numerous small white stars and several stylized planets. In the top left, there's a planet with blue and white stripes. Below it, a planet with orange and white stripes. In the bottom right, a large planet with yellow and red stripes is surrounded by a ring system. Other smaller planets with various patterns and colors are scattered across the scene.

02

Modélisation

Agents, objets et environnement

- Agents
 - Pays (countries)
- Objets :
 - Astres (Stars)
 - Planete (planet)
 - Satellite
 - Fusée (Rocket)
- Environnement
 - Système solaire (Solar System)



The background is a vibrant, stylized space scene. It features a gradient of red, purple, and blue. There are several planets: one with blue and white stripes in the top left, one with orange and white stripes in the top right, and a large one with yellow and red stripes and a red ring in the bottom right. Numerous small white stars and dots are scattered throughout the background.

03

Simulation

Fonctionnement des agents

Percept :

- Les agents observent l'environnement, listent les planètes qu'ils peuvent atteindre (ressources disponibles, acquises ou non)

Deliberate (quelle action choisir ?) :

- Observer une planète
- Préparer l'équipage au lancement
- Lancer une fusée
- Abandonner un projet de lancement
- Demander / Accepter une alliance

Act :

- Réalisation de l'action précédemment choisie

Communication et synchronisation

- Multi-threading avec goroutine pour les agents et fusées
- Mutex
- WaitGroup
- Channel

Des stratégies d'alliance et plus

Plusieurs type de stratégies :

- Stratégies d'observation
- Stratégies de préparation
- Stratégies de changement
- Stratégies de réaction
- Stratégies d'alliances

Fonctionnement des alliances

Création des alliances :

- Stratégie d'alliances
- Modèle de communication "message-passing"

Utilisation des alliances :

- Aide financière au lancement de fusée

Récompense des alliances :

- Territoires
- Ressources



04

Interface

FRONTEND: Technologies



React.JS

Librairie Javascript



Typescript

Extension de Javascript
typée



ReactThreeFiber

Librairie basé sur
Three.js pour la 3D



CONFIG

SHUFFLE

START

	FRANCE	937056
	NORTH KOREA	35978
	SOUTH KOREA	864965
	UNITED STATES	260449
	UNITED KINGDOM	938154
	INDIA	178232
	CHINA	902208
	RUSSIA	795103
	CENTRAL AFRICA	659620
	AUSTRALIA	42235
	ITALY	238899
	GERMANY	504890
	MEXICO	775929
	CANADA	738938
	JAPAN	940445



COUNTRIES

 mexico | russia | south_korea | india | united_states | china | australia | central_africa

 venus | io | titan | europa

 **MEXICO** preparing the launch to Neptune. . +

 **CANADA** preparing the launch to Uranus. . +

 **ITALY** preparing the launch to Uranus. . -

 65520 |  0

 mexico | south_korea | india | australia | japan

 europa

 **FRANCE** waiting rocket to arrived at Jupiter. . +

 **RUSSIA** preparing the launch to Neptune. . +

 **AUSTRALIA** waiting rocket to arrived at Uranus. . -

 34322 |  566

 italy | united_states | china



 x: 1020715 | y: 0.00 | z: -204732

 **CENTRAL AFRICA** preparing the launch to Jupiter. . -

 30571 |  329

 south_korea | north_korea

 mercury

FRONTEND



	UNITED STATES waiting rocket to arrived at Uranus...	+
	UNITED KINGDOM job done.	+
	CENTRAL AFRICA job done.	-
 54570  347		
 france north_korea russia china australia		
 ganymede callisto triton europa venus		
	FRANCE job done.	+
	INDIA job done.	+
	CHINA alliance with Russia...	+

JUPITER

x



4951 ~ 5455 distance from The Sun

x: 369782 | y: 0.00 | z: -3769.27

0.01 | 28.70 | 0.00 | 10485

lo | europa | ganymede | callisto

india

COUNTRIES

waiting rocket to arrived at Uranus...

SOUTH KOREA
waiting rocket to arrived at Pluto...

NORTH KOREA
preparing the launch to Pluto...

RUSSIA
preparing the launch to Uranus...

INDIA
preparing the launch to Pluto...

CHINA
preparing the launch to Uranus...

GERMANY
waiting rocket to arrived at Pluto...

ITALY
preparing the launch to Pluto...

MEXICO
observing Saturn...

UNITED STATES
waiting rocket to arrived at Pluto...

UNITED KINGDOM
preparing the launch to Uranus...

948905

0

people

globe

callisto

JAPAN
waiting rocket to arrived at Uranus ..

931699

589

people

globe

triton

europa

x: 12983.57

y: 0.00

z: 5175.88

FRANCE
preparing the launch to Pluto ..

JUPITER

x



4951 ~ 5455 distance from The Sun

x: 4873.40 | y: 0.00 | z: 2424.20

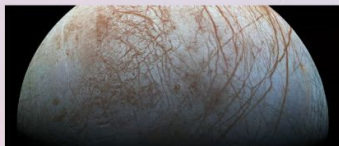
↶ 0.01 | ↷ 28.70 | ↻ 0.00 | 💎 9

☾ lo | europa | ganymede | callisto

🚩 India

EUROPA

x



35 distance from Jupiter

x: 4873.40 | y: 0.00 | z: 2424.20

↶ 0.01 | ↷ 28.70 | ↻ 0.00 | 💎 3

🚩 japan

Callisto
Ganymede

Io

COUNTRIES

	CENTRAL AFRICA job done.	+
	AUSTRALIA job done.	+
	CANADA job done.	+
	SOUTH KOREA job done.	+
	NORTH KOREA job done.	+
	RUSSIA job done.	+
	INDIA job done.	+
	CHINA job done.	+
	GERMANY job done.	+
	ITALY job done.	+
	MEXICO job done.	+
	UNITED STATES job done.	+
	UNITED KINGDOM job done.	+
	JAPAN job done.	+
	FRANCE job done.	+

SUMMARY



CENTRAL AFRICA

planets: 4

territory: 0.8/15 (5.33%)

alliances: 3

strategy:

observation: if_lo_slow

alliance: always

skipping: if_hi

launch: if_hi_medium



CHINA

planets: 6

territory: 2.3/15 (15.33%)

alliances: 1

strategy:

observation: if_hi_quick

alliance: if_hi

skipping: always

launch: if_lo_slow

Forces

- Stratégie visible
- Comportement inattendu
- Non déterministe
- Multi threading : les agents sont indépendants
- Communication entre agent

Faiblesses

- Peu de configuration : beaucoup d'aléatoire par manque de temps
- Monde restreint : l'échelle est faible
- Décision stratégique moyennement complexe : intelligence moyenne

Amélioration

- Plus de configuration
- Plus de systèmes solaires
- Espionnage et sabotage : géopolitique
- Système plus avancé d'intelligence et de stratégie

Stratégie prometteuse

Avec des ressources aléatoirement distribuées sur les planètes :

- Observation rapide de l'environnement
- Préparation lente du lancement des fusées



**Merci de votre
attention !**

